Aggregate Demand Curve and Solution of the Dynamic Model

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Aggregate Demand curve.

$$y = [1+\tau(\theta)] \times \overset{\text{Euler equation}}{c} = 1+\tau(\theta) \times \left[\frac{\delta - r}{\sigma'(\theta)}\right]^{\varepsilon} \times \frac{1}{[1+\tau(\theta)]^{\varepsilon}}$$

$$\boxed{y = \left[\frac{\delta - r}{\sigma'(\theta)}\right]^{\varepsilon} \times \frac{1}{[1+\tau(\theta)]^{\varepsilon - 1}} = y^d(\theta)}$$

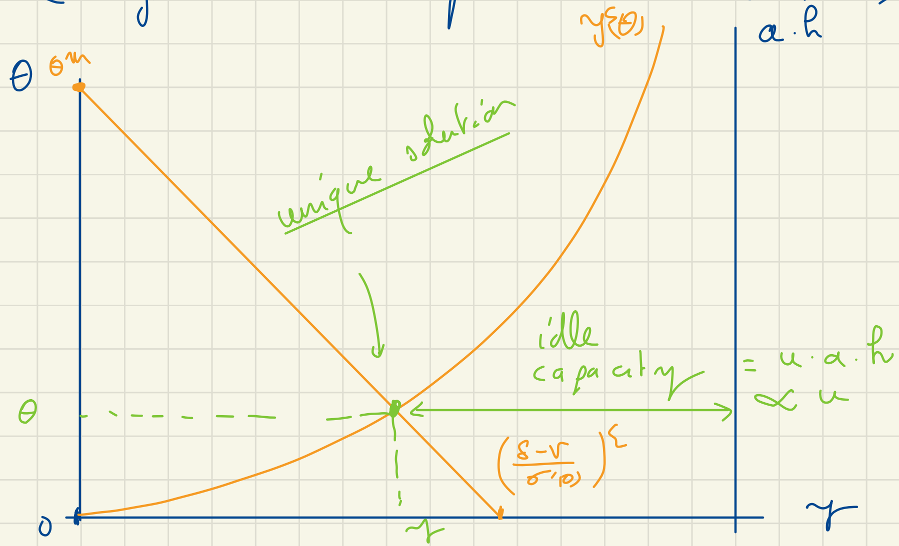
- $y^d(0) = \left[\frac{\delta - r}{\sigma'(0)}\right]^{\varepsilon} > 0$
- $y^d(\theta_m) = 0$
- $y^d(\theta)$ is $\searrow$ with θ

Solution of model:

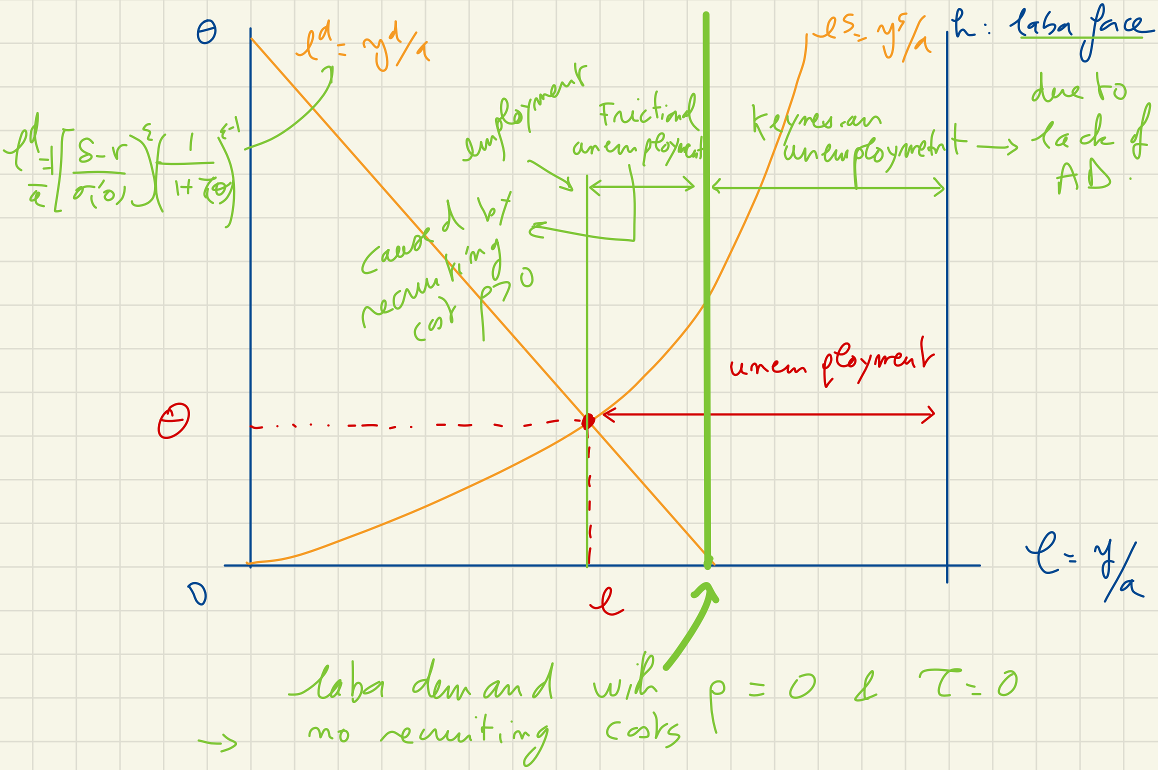
1. θ satisfies $\boxed{\underbrace{\overset{\text{AD}}{y^d(\theta)}}_{\text{Euler}} = \underbrace{\overset{\text{AS}}{y^s(\theta)}}_{\text{beveridge}}}$

2. from θ: compute all other variables

Finding θ:



Decomposition of unemployment: Keynesian + frictional



With low AD:
- Keynesian unemployment ↗
- Frictional unemployment ↘
- Total unemployment ↗

With high AD:
- Keynesian unemployment ↘
- Frictional unemployment ↗
- Total unemployment ↘

See Michaillat (2012)